Deep Learning Portfolio Examination

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Part - 1: Stabilizing the Existing Codebase

Performance Report

Each of the four recovered diagnostic image profiles (cells, chest, lesions, orgs) was benchmarked against all three architectures (AlexNet, VGG16, ResNet18) using the reconstructed, config-driven pipeline. The complete per-epoch training and validation logs and the final held-out test metrics are reproduced below for every dataset-model permutation.

Dataset Used – cells

Blood cell microscopy (8 classes, RGB) | Expected minimum test accuracy: 90%

1. cells – AlexNet

Total epochs trained: 15

Configuration Used:

```
ACTIVATION.................. ReLU
BATCH_SIZE.................. 64
CHANNELS.................... 3
CRITERION................... CrossEntropyLoss
DATA_PATH................... ../data
DEVICE...................... cuda
DROP_RATE................... 0.5
EPOCHS...................... 15
LEARNING_RATE............... 0.001
NUM_CLASSES................. 8
OPTIMIZER................... Adam
RANDOM_SEED................. 42
VAL_SPLIT................... 0.1
WEIGHT_DECAY................ 0
```

Logs:

```
Epoch [01/15] | Train Loss: 0.9301 - Train Acc: 63.56% | Val Loss: 3.1397 - Val Acc: 36.36%
Epoch [02/15] | Train Loss: 0.4248 - Train Acc: 84.58% | Val Loss: 0.9293 - Val Acc: 68.25%
Epoch [03/15] | Train Loss: 0.3071 - Train Acc: 89.87% | Val Loss: 0.3447 - Val Acc: 88.59%
Epoch [04/15] | Train Loss: 0.2565 - Train Acc: 91.68% | Val Loss: 0.5073 - Val Acc: 79.81%
Epoch [05/15] | Train Loss: 0.2163 - Train Acc: 93.25% | Val Loss: 2.3361 - Val Acc: 57.94%
Epoch [06/15] | Train Loss: 0.1875 - Train Acc: 93.78% | Val Loss: 0.4023 - Val Acc: 86.32%
Epoch [07/15] | Train Loss: 0.1736 - Train Acc: 94.26% | Val Loss: 1.6407 - Val Acc: 64.01%
Epoch [08/15] | Train Loss: 0.1862 - Train Acc: 94.21% | Val Loss: 0.8748 - Val Acc: 69.86%
Epoch [09/15] | Train Loss: 0.1644 - Train Acc: 94.78% | Val Loss: 0.2744 - Val Acc: 89.83%
Epoch [10/15] | Train Loss: 0.1454 - Train Acc: 95.27% | Val Loss: 0.2365 - Val Acc: 92.90%
Epoch [11/15] | Train Loss: 0.1482 - Train Acc: 95.17% | Val Loss: 0.2147 - Val Acc: 93.12%
Epoch [12/15] | Train Loss: 0.1422 - Train Acc: 95.55% | Val Loss: 0.2316 - Val Acc: 92.47%
Epoch [13/15] | Train Loss: 0.1256 - Train Acc: 95.94% | Val Loss: 0.3447 - Val Acc: 91.15%
Epoch [14/15] | Train Loss: 0.1046 - Train Acc: 96.76% | Val Loss: 0.2586 - Val Acc: 92.90%
Epoch [15/15] | Train Loss: 0.0985 - Train Acc: 96.85% | Val Loss: 0.2213 - Val Acc: 93.64%
Best checkpoint saved: best_model/cells_AlexNet.pth (epoch=15, val_acc=93.64%)
Training Complete!
```

Final Test Results (AlexNet on cells): Accuracy = 95.15% | Precision (macro) = 0.9474 | Recall (macro) = 0.9443 | Macro F1 = 0.9445

2. cells – VGG16

Total epochs trained: 15

Configuration Used:

```
ACTIVATION.................. ReLU
BATCH_SIZE.................. 64
CHANNELS.................... 3
CRITERION................... CrossEntropyLoss
DATA_PATH................... ../data
DEVICE...................... cuda
DROP_RATE................... 0.5
EPOCHS...................... 15
LEARNING_RATE............... 0.001
NUM_CLASSES................. 8
OPTIMIZER................... Adam
```

```
RANDOM_SEED.................. 42
VAL_SPLIT.................... 0.1
WEIGHT_DECAY................. 0
```

Logs:

```
Epoch [01/15] | Train Loss: 0.7807 - Train Acc: 70.35% | Val Loss: 0.5020 - Val Acc: 81.57%
Epoch [02/15] | Train Loss: 0.4778 - Train Acc: 83.65% | Val Loss: 0.7367 - Val Acc: 78.05%
Epoch [03/15] | Train Loss: 0.3864 - Train Acc: 87.65% | Val Loss: 0.6123 - Val Acc: 77.76%
Epoch [04/15] | Train Loss: 0.3115 - Train Acc: 90.32% | Val Loss: 0.6647 - Val Acc: 75.20%
Epoch [05/15] | Train Loss: 0.3170 - Train Acc: 90.10% | Val Loss: 0.6265 - Val Acc: 80.03%
Epoch [06/15] | Train Loss: 0.2479 - Train Acc: 92.34% | Val Loss: 0.3569 - Val Acc: 89.39%
Epoch [07/15] | Train Loss: 0.2155 - Train Acc: 93.08% | Val Loss: 0.2055 - Val Acc: 93.93%
Epoch [08/15] | Train Loss: 0.2004 - Train Acc: 93.95% | Val Loss: 0.4448 - Val Acc: 86.83%
Epoch [09/15] | Train Loss: 0.1843 - Train Acc: 94.57% | Val Loss: 0.2220 - Val Acc: 93.05%
Epoch [10/15] | Train Loss: 0.1573 - Train Acc: 95.13% | Val Loss: 0.1483 - Val Acc: 95.68%
Epoch [11/15] | Train Loss: 0.1366 - Train Acc: 95.77% | Val Loss: 0.3248 - Val Acc: 88.66%
Epoch [12/15] | Train Loss: 0.1574 - Train Acc: 95.20% | Val Loss: 0.1593 - Val Acc: 94.51%
Epoch [13/15] | Train Loss: 0.1475 - Train Acc: 95.42% | Val Loss: 0.1970 - Val Acc: 93.34%
Epoch [14/15] | Train Loss: 0.1287 - Train Acc: 95.91% | Val Loss: 0.1637 - Val Acc: 95.25%
Epoch [15/15] | Train Loss: 0.1112 - Train Acc: 96.65% | Val Loss: 0.1890 - Val Acc: 94.15%
Best checkpoint saved: best_model/cells_VGG16.pth (epoch=10, val_acc=95.68%)
Training Complete!
```

Final Test Results (VGG16 on cells): Accuracy = 96.78% | Precision (macro) = 0.9646 | Recall (macro) = 0.9685 | Macro F1 = 0.9665

3. cells – ResNet18

Total epochs trained: 15

Configuration Used:

```
ACTIVATION................... ReLU
BATCH_SIZE................... 64
CHANNELS..................... 3
CRITERION.................... CrossEntropyLoss
DATA_PATH.................... ../data
DEVICE....................... cuda
DROP_RATE.................... 0.5
EPOCHS....................... 15
LEARNING_RATE................ 0.001
NUM_CLASSES.................. 8
OPTIMIZER.................... Adam
RANDOM_SEED.................. 42
VAL_SPLIT.................... 0.1
WEIGHT_DECAY................. 0
```

Logs:

```
Epoch [01/15] | Train Loss: 0.4373 - Train Acc: 84.42% | Val Loss: 0.7451 - Val Acc: 79.44%
Epoch [02/15] | Train Loss: 0.2135 - Train Acc: 92.99% | Val Loss: 1.0535 - Val Acc: 73.45%
Epoch [03/15] | Train Loss: 0.1472 - Train Acc: 95.23% | Val Loss: 0.4059 - Val Acc: 85.52%
Epoch [04/15] | Train Loss: 0.1450 - Train Acc: 95.02% | Val Loss: 0.2614 - Val Acc: 90.71%
Epoch [05/15] | Train Loss: 0.1137 - Train Acc: 96.25% | Val Loss: 0.7099 - Val Acc: 80.61%
Epoch [06/15] | Train Loss: 0.1011 - Train Acc: 96.54% | Val Loss: 0.2834 - Val Acc: 90.86%
Epoch [07/15] | Train Loss: 0.1131 - Train Acc: 96.21% | Val Loss: 0.2845 - Val Acc: 90.56%
Epoch [08/15] | Train Loss: 0.0874 - Train Acc: 97.11% | Val Loss: 0.3889 - Val Acc: 88.88%
Epoch [09/15] | Train Loss: 0.0906 - Train Acc: 96.68% | Val Loss: 0.4331 - Val Acc: 86.47%
Epoch [10/15] | Train Loss: 0.0812 - Train Acc: 97.24% | Val Loss: 0.3649 - Val Acc: 88.95%
Epoch [11/15] | Train Loss: 0.0669 - Train Acc: 97.71% | Val Loss: 0.5524 - Val Acc: 86.10%
Epoch [12/15] | Train Loss: 0.0813 - Train Acc: 97.22% | Val Loss: 0.5019 - Val Acc: 85.81%
Epoch [13/15] | Train Loss: 0.0628 - Train Acc: 97.85% | Val Loss: 0.6183 - Val Acc: 85.59%
Epoch [14/15] | Train Loss: 0.0668 - Train Acc: 97.74% | Val Loss: 0.2208 - Val Acc: 93.20%
Epoch [15/15] | Train Loss: 0.0569 - Train Acc: 98.07% | Val Loss: 0.1266 - Val Acc: 96.12%
Best checkpoint saved: best_model/cells_ResNet18.pth (epoch=15, val_acc=96.12%)
Training Complete!
```

Final Test Results (ResNet18 on cells): Accuracy = 96.35% | Precision (macro) = 0.9599 | Recall (macro) = 0.9641 | Macro F1 = 0.9613

Dataset Used – chest

Chest X-ray (2 classes, grayscale) | Expected minimum test accuracy: 87%

4. chest – AlexNet

Total epochs trained: 15

Configuration Used:

```
ACTIVATION................... ReLU
BATCH_SIZE................... 64
CHANNELS..................... 1
```

```
CRITERION.................... CrossEntropyLoss
DATA_PATH.................... ../data
DEVICE....................... cuda
DROP_RATE.................... 0.5
EPOCHS....................... 15
LEARNING_RATE................ 0.001
NUM_CLASSES.................. 2
OPTIMIZER.................... Adam
RANDOM_SEED.................. 42
VAL_SPLIT.................... 0.1
WEIGHT_DECAY................. 1e-4
```

Logs:

```
Epoch [01/15] | Train Loss: 0.3124 - Train Acc: 87.15% | Val Loss: 0.2966 - Val Acc: 91.97%
Epoch [02/15] | Train Loss: 0.1586 - Train Acc: 94.20% | Val Loss: 0.2135 - Val Acc: 94.84%
Epoch [03/15] | Train Loss: 0.1320 - Train Acc: 94.84% | Val Loss: 0.1117 - Val Acc: 95.22%
Epoch [04/15] | Train Loss: 0.1228 - Train Acc: 95.63% | Val Loss: 0.1242 - Val Acc: 95.79%
Epoch [05/15] | Train Loss: 0.0795 - Train Acc: 97.01% | Val Loss: 0.4370 - Val Acc: 86.04%
Epoch [06/15] | Train Loss: 0.0965 - Train Acc: 96.56% | Val Loss: 0.3662 - Val Acc: 86.62%
Epoch [07/15] | Train Loss: 0.0977 - Train Acc: 96.58% | Val Loss: 0.1532 - Val Acc: 95.03%
Epoch [08/15] | Train Loss: 0.0780 - Train Acc: 97.41% | Val Loss: 0.1836 - Val Acc: 95.79%
Epoch [09/15] | Train Loss: 0.0614 - Train Acc: 97.88% | Val Loss: 0.1328 - Val Acc: 95.98%
Epoch [10/15] | Train Loss: 0.0595 - Train Acc: 97.90% | Val Loss: 0.1617 - Val Acc: 95.03%
Epoch [11/15] | Train Loss: 0.0482 - Train Acc: 98.19% | Val Loss: 0.1559 - Val Acc: 95.22%
Epoch [12/15] | Train Loss: 0.0465 - Train Acc: 98.45% | Val Loss: 0.0932 - Val Acc: 96.75%
Epoch [13/15] | Train Loss: 0.0471 - Train Acc: 98.51% | Val Loss: 0.1422 - Val Acc: 95.98%
Epoch [14/15] | Train Loss: 0.0393 - Train Acc: 98.47% | Val Loss: 0.0817 - Val Acc: 97.90%
Epoch [15/15] | Train Loss: 0.0348 - Train Acc: 98.92% | Val Loss: 0.1391 - Val Acc: 95.22%
Best checkpoint saved: best_model/chest_AlexNet.pth (epoch=14, val_acc=97.90%)
Training Complete!
```

Final Test Results (AlexNet on chest): Accuracy = 86.38% | Precision (macro) = 0.9054 | Recall (macro) = 0.8201 | Macro F1 = 0.8408

5. chest – VGG16

Total epochs trained: 15

Configuration Used:

```
ACTIVATION................... ReLU
BATCH_SIZE................... 64
CHANNELS..................... 1
CRITERION.................... CrossEntropyLoss
DATA_PATH.................... ../data
DEVICE....................... cuda
DROP_RATE.................... 0.5
EPOCHS....................... 15
LEARNING_RATE................ 0.001
NUM_CLASSES.................. 2
OPTIMIZER.................... Adam
RANDOM_SEED.................. 42
VAL_SPLIT.................... 0.1
WEIGHT_DECAY................. 1e-4
```

Logs:

```
Epoch [01/15] | Train Loss: 0.2293 - Train Acc: 90.59% | Val Loss: 0.3932 - Val Acc: 85.85%
Epoch [02/15] | Train Loss: 0.1342 - Train Acc: 95.29% | Val Loss: 0.5343 - Val Acc: 83.17%
Epoch [03/15] | Train Loss: 0.0983 - Train Acc: 96.45% | Val Loss: 0.1144 - Val Acc: 95.22%
Epoch [04/15] | Train Loss: 0.0907 - Train Acc: 96.58% | Val Loss: 0.1326 - Val Acc: 95.60%
Epoch [05/15] | Train Loss: 0.0853 - Train Acc: 96.62% | Val Loss: 1.8916 - Val Acc: 61.19%
Epoch [06/15] | Train Loss: 0.1042 - Train Acc: 96.50% | Val Loss: 0.1755 - Val Acc: 95.98%
Epoch [07/15] | Train Loss: 0.0813 - Train Acc: 97.15% | Val Loss: 0.1425 - Val Acc: 94.65%
Epoch [08/15] | Train Loss: 0.0695 - Train Acc: 97.49% | Val Loss: 0.0914 - Val Acc: 95.60%
Epoch [09/15] | Train Loss: 0.0548 - Train Acc: 97.98% | Val Loss: 0.2228 - Val Acc: 95.03%
Epoch [10/15] | Train Loss: 0.0496 - Train Acc: 98.15% | Val Loss: 0.0964 - Val Acc: 95.98%
Epoch [11/15] | Train Loss: 0.0610 - Train Acc: 97.90% | Val Loss: 0.3047 - Val Acc: 91.20%
Epoch [12/15] | Train Loss: 0.0487 - Train Acc: 98.13% | Val Loss: 0.1201 - Val Acc: 96.18%
Epoch [13/15] | Train Loss: 0.0488 - Train Acc: 98.24% | Val Loss: 0.1335 - Val Acc: 95.98%
Epoch [14/15] | Train Loss: 0.0437 - Train Acc: 98.81% | Val Loss: 0.2581 - Val Acc: 95.98%
Epoch [15/15] | Train Loss: 0.0345 - Train Acc: 98.68% | Val Loss: 0.0953 - Val Acc: 96.94%
Best checkpoint saved: best_model/chest_VGG16.pth (epoch=15, val_acc=96.94%)
Training Complete!
```

Final Test Results (VGG16 on chest): Accuracy = 82.21% | Precision (macro) = 0.8892 | Recall (macro) = 0.7628 | Macro F1 = 0.7822

6. chest – ResNet18

Total epochs trained: 15

Configuration Used:

```
ACTIVATION.................. ReLU
BATCH_SIZE.................. 64
CHANNELS.................... 1
CRITERION................... CrossEntropyLoss
DATA_PATH................... ../data
DEVICE...................... cuda
DROP_RATE................... 0.5
EPOCHS...................... 15
LEARNING_RATE............... 0.001
NUM_CLASSES................. 2
OPTIMIZER................... Adam
RANDOM_SEED................. 42
VAL_SPLIT................... 0.1
WEIGHT_DECAY................ 1e-4
```

Logs:

```
Epoch [01/15] | Train Loss: 0.2396 - Train Acc: 90.25% | Val Loss: 0.3194 - Val Acc: 87.95%
Epoch [02/15] | Train Loss: 0.1516 - Train Acc: 94.18% | Val Loss: 0.4163 - Val Acc: 86.81%
Epoch [03/15] | Train Loss: 0.1176 - Train Acc: 95.52% | Val Loss: 0.2554 - Val Acc: 91.59%
Epoch [04/15] | Train Loss: 0.1124 - Train Acc: 95.82% | Val Loss: 0.1806 - Val Acc: 93.69%
Epoch [05/15] | Train Loss: 0.0987 - Train Acc: 96.24% | Val Loss: 1.6548 - Val Acc: 65.97%
Epoch [06/15] | Train Loss: 0.0926 - Train Acc: 96.64% | Val Loss: 0.2739 - Val Acc: 91.01%
Epoch [07/15] | Train Loss: 0.0781 - Train Acc: 97.11% | Val Loss: 0.1037 - Val Acc: 95.98%
Epoch [08/15] | Train Loss: 0.0702 - Train Acc: 97.52% | Val Loss: 0.1801 - Val Acc: 93.69%
Epoch [09/15] | Train Loss: 0.0729 - Train Acc: 97.09% | Val Loss: 0.1435 - Val Acc: 93.88%
Epoch [10/15] | Train Loss: 0.0638 - Train Acc: 97.83% | Val Loss: 0.1197 - Val Acc: 95.79%
Epoch [11/15] | Train Loss: 0.0601 - Train Acc: 97.81% | Val Loss: 0.2391 - Val Acc: 92.35%
Epoch [12/15] | Train Loss: 0.0569 - Train Acc: 97.81% | Val Loss: 0.1293 - Val Acc: 95.60%
Epoch [13/15] | Train Loss: 0.0540 - Train Acc: 98.00% | Val Loss: 0.2655 - Val Acc: 91.59%
Epoch [14/15] | Train Loss: 0.0507 - Train Acc: 98.07% | Val Loss: 0.7436 - Val Acc: 81.64%
Epoch [15/15] | Train Loss: 0.0358 - Train Acc: 98.68% | Val Loss: 1.1647 - Val Acc: 78.20%
Best checkpoint saved: best_model/chest_ResNet18.pth (epoch=7, val_acc=95.98%)
Training Complete!
```

Final Test Results (ResNet18 on chest): Accuracy = 84.46% | Precision (macro) = 0.8866 | Recall (macro) = 0.7970 | Macro F1 = 0.8168

Dataset Used – lesions

Skin lesion dermatoscopy (7 classes, RGB) | Expected minimum test accuracy: 67%

7. lesions – AlexNet

Total epochs trained: 25

Configuration Used:

```
ACTIVATION.................. ReLU
BATCH_SIZE.................. 64
CHANNELS.................... 3
CRITERION................... CrossEntropyLoss
DATA_PATH................... ../data
DEVICE...................... cuda
DROP_RATE................... 0.5
EPOCHS...................... 25
LEARNING_RATE............... 0.001
NUM_CLASSES................. 7
OPTIMIZER................... Adam
RANDOM_SEED................. 42
VAL_SPLIT................... 0.1
WEIGHT_DECAY................ 1e-4
```

Logs:

```
Epoch [01/25] | Train Loss: 1.0529 - Train Acc: 65.61% | Val Loss: 1.1957 - Val Acc: 70.16%
Epoch [02/25] | Train Loss: 0.8855 - Train Acc: 67.90% | Val Loss: 0.8012 - Val Acc: 70.41%
Epoch [03/25] | Train Loss: 0.8424 - Train Acc: 70.00% | Val Loss: 0.7480 - Val Acc: 73.41%
Epoch [04/25] | Train Loss: 0.8165 - Train Acc: 70.72% | Val Loss: 0.9198 - Val Acc: 63.05%
Epoch [05/25] | Train Loss: 0.8063 - Train Acc: 70.76% | Val Loss: 0.8240 - Val Acc: 71.16%
Epoch [06/25] | Train Loss: 0.7798 - Train Acc: 72.01% | Val Loss: 0.6731 - Val Acc: 75.53%
Epoch [07/25] | Train Loss: 0.7447 - Train Acc: 72.62% | Val Loss: 0.6804 - Val Acc: 74.78%
Epoch [08/25] | Train Loss: 0.7338 - Train Acc: 73.69% | Val Loss: 0.6499 - Val Acc: 75.91%
Epoch [09/25] | Train Loss: 0.7204 - Train Acc: 73.98% | Val Loss: 0.6923 - Val Acc: 74.66%
Epoch [10/25] | Train Loss: 0.7095 - Train Acc: 74.60% | Val Loss: 0.6509 - Val Acc: 76.78%
Epoch [11/25] | Train Loss: 0.6972 - Train Acc: 74.38% | Val Loss: 0.7455 - Val Acc: 73.41%
Epoch [12/25] | Train Loss: 0.6763 - Train Acc: 75.61% | Val Loss: 0.6667 - Val Acc: 75.28%
Epoch [13/25] | Train Loss: 0.6642 - Train Acc: 75.68% | Val Loss: 0.8356 - Val Acc: 72.53%
Epoch [14/25] | Train Loss: 0.6372 - Train Acc: 76.40% | Val Loss: 0.9236 - Val Acc: 60.67%
```

```
Epoch [15/25] | Train Loss: 0.6312 - Train Acc: 76.95% | Val Loss: 1.0745 - Val Acc: 55.93%
Epoch [16/25] | Train Loss: 0.6127 - Train Acc: 77.20% | Val Loss: 0.6772 - Val Acc: 75.28%
Epoch [17/25] | Train Loss: 0.6067 - Train Acc: 77.76% | Val Loss: 0.6645 - Val Acc: 76.65%
Epoch [18/25] | Train Loss: 0.5895 - Train Acc: 78.18% | Val Loss: 0.6799 - Val Acc: 74.28%
Epoch [19/25] | Train Loss: 0.5719 - Train Acc: 78.68% | Val Loss: 0.9693 - Val Acc: 72.53%
Epoch [20/25] | Train Loss: 0.5632 - Train Acc: 79.40% | Val Loss: 0.6156 - Val Acc: 78.28%
Epoch [21/25] | Train Loss: 0.5479 - Train Acc: 80.08% | Val Loss: 0.6048 - Val Acc: 78.78%
Epoch [22/25] | Train Loss: 0.5367 - Train Acc: 80.25% | Val Loss: 1.0047 - Val Acc: 74.78%
Epoch [23/25] | Train Loss: 0.5145 - Train Acc: 81.59% | Val Loss: 1.0500 - Val Acc: 62.92%
Epoch [24/25] | Train Loss: 0.5147 - Train Acc: 81.05% | Val Loss: 1.1849 - Val Acc: 58.55%
Epoch [25/25] | Train Loss: 0.5011 - Train Acc: 81.45% | Val Loss: 0.6626 - Val Acc: 77.15%
Best checkpoint saved: best_model/lesions_AlexNet.pth (epoch=21, val_acc=78.78%)
Training Complete!
```

Final Test Results (AlexNet on lesions): Accuracy = 76.01% | Precision (macro) = 0.5651 | Recall (macro) = 0.4770 | Macro F1 = 0.4786

8. lesions – VGG16

Total epochs trained: 25

Configuration Used:

```
ACTIVATION.................. ReLU
BATCH_SIZE.................. 64
CHANNELS.................... 3
CRITERION................... CrossEntropyLoss
DATA_PATH................... ../data
DEVICE...................... cuda
DROP_RATE................... 0.5
EPOCHS...................... 25
LEARNING_RATE............... 0.001
NUM_CLASSES................. 7
OPTIMIZER................... Adam
RANDOM_SEED................. 42
VAL_SPLIT................... 0.1
WEIGHT_DECAY................ 1e-4
```

Logs:

```
Epoch [01/25] | Train Loss: 1.0747 - Train Acc: 65.63% | Val Loss: 0.9127 - Val Acc: 70.16%
Epoch [02/25] | Train Loss: 0.9660 - Train Acc: 66.60% | Val Loss: 0.8620 - Val Acc: 69.29%
Epoch [03/25] | Train Loss: 0.9424 - Train Acc: 66.32% | Val Loss: 0.8262 - Val Acc: 68.54%
Epoch [04/25] | Train Loss: 0.9072 - Train Acc: 67.01% | Val Loss: 0.8024 - Val Acc: 70.79%
Epoch [05/25] | Train Loss: 0.9077 - Train Acc: 66.83% | Val Loss: 0.7764 - Val Acc: 71.66%
Epoch [06/25] | Train Loss: 0.8819 - Train Acc: 67.62% | Val Loss: 0.7903 - Val Acc: 71.29%
Epoch [07/25] | Train Loss: 0.8882 - Train Acc: 67.43% | Val Loss: 0.8117 - Val Acc: 69.29%
Epoch [08/25] | Train Loss: 0.8679 - Train Acc: 68.01% | Val Loss: 0.7641 - Val Acc: 71.91%
Epoch [09/25] | Train Loss: 0.8726 - Train Acc: 68.37% | Val Loss: 0.8375 - Val Acc: 67.04%
Epoch [10/25] | Train Loss: 0.8446 - Train Acc: 69.64% | Val Loss: 0.7748 - Val Acc: 72.78%
Epoch [11/25] | Train Loss: 0.8310 - Train Acc: 69.57% | Val Loss: 0.7272 - Val Acc: 74.28%
Epoch [12/25] | Train Loss: 0.8144 - Train Acc: 69.82% | Val Loss: 0.7338 - Val Acc: 73.53%
Epoch [13/25] | Train Loss: 0.8125 - Train Acc: 70.27% | Val Loss: 0.7923 - Val Acc: 70.29%
Epoch [14/25] | Train Loss: 0.8022 - Train Acc: 70.50% | Val Loss: 0.8496 - Val Acc: 69.04%
Epoch [15/25] | Train Loss: 0.7924 - Train Acc: 70.61% | Val Loss: 0.7144 - Val Acc: 72.91%
Epoch [16/25] | Train Loss: 0.7932 - Train Acc: 70.61% | Val Loss: 0.9979 - Val Acc: 62.17%
Epoch [17/25] | Train Loss: 0.7909 - Train Acc: 70.76% | Val Loss: 0.6833 - Val Acc: 73.66%
Epoch [18/25] | Train Loss: 0.7714 - Train Acc: 71.47% | Val Loss: 0.7171 - Val Acc: 72.28%
Epoch [19/25] | Train Loss: 0.7749 - Train Acc: 71.79% | Val Loss: 0.7009 - Val Acc: 75.16%
Epoch [20/25] | Train Loss: 0.7584 - Train Acc: 71.83% | Val Loss: 1.5402 - Val Acc: 56.05%
Epoch [21/25] | Train Loss: 0.7425 - Train Acc: 73.23% | Val Loss: 0.8270 - Val Acc: 69.79%
Epoch [22/25] | Train Loss: 0.7476 - Train Acc: 72.51% | Val Loss: 0.7117 - Val Acc: 73.16%
Epoch [23/25] | Train Loss: 0.7246 - Train Acc: 73.10% | Val Loss: 0.8416 - Val Acc: 74.41%
Epoch [24/25] | Train Loss: 0.7221 - Train Acc: 73.57% | Val Loss: 0.6949 - Val Acc: 75.03%
Epoch [25/25] | Train Loss: 0.7133 - Train Acc: 74.27% | Val Loss: 0.8404 - Val Acc: 67.29%
Best checkpoint saved: best_model/lesions_VGG16.pth (epoch=19, val_acc=75.16%)
Training Complete!
[sklearn] UndefinedMetricWarning: precision ill-defined (some classes had 0 predicted samples).
```

Final Test Results (VGG16 on lesions): Accuracy = 71.22% | Precision (macro) = 0.3895 | Recall (macro) = 0.3053 | Macro F1 = 0.3239

9. lesions – ResNet18

Total epochs trained: 25

Configuration Used:

```
ACTIVATION.................. ReLU
BATCH_SIZE.................. 64
CHANNELS.................... 3
CRITERION................... CrossEntropyLoss
DATA_PATH................... ../data
```

```
DEVICE...................... cuda
DROP_RATE................... 0.5
EPOCHS...................... 25
LEARNING_RATE............... 0.001
NUM_CLASSES................. 7
OPTIMIZER................... Adam
RANDOM_SEED................. 42
VAL_SPLIT................... 0.1
WEIGHT_DECAY................ 1e-4
```

Logs:

```
Epoch [01/25] | Train Loss: 0.9720 - Train Acc: 66.03% | Val Loss: 0.9265 - Val Acc: 68.41%
Epoch [02/25] | Train Loss: 0.8537 - Train Acc: 68.29% | Val Loss: 0.7988 - Val Acc: 71.79%
Epoch [03/25] | Train Loss: 0.8348 - Train Acc: 69.14% | Val Loss: 0.7086 - Val Acc: 74.03%
Epoch [04/25] | Train Loss: 0.8075 - Train Acc: 70.56% | Val Loss: 0.8540 - Val Acc: 63.92%
Epoch [05/25] | Train Loss: 0.7779 - Train Acc: 70.69% | Val Loss: 0.7274 - Val Acc: 72.53%
Epoch [06/25] | Train Loss: 0.7519 - Train Acc: 72.01% | Val Loss: 0.7053 - Val Acc: 73.16%
Epoch [07/25] | Train Loss: 0.7354 - Train Acc: 72.51% | Val Loss: 0.7191 - Val Acc: 72.16%
Epoch [08/25] | Train Loss: 0.7096 - Train Acc: 73.64% | Val Loss: 0.9580 - Val Acc: 67.29%
Epoch [09/25] | Train Loss: 0.7143 - Train Acc: 73.24% | Val Loss: 0.7481 - Val Acc: 73.03%
Epoch [10/25] | Train Loss: 0.6814 - Train Acc: 74.85% | Val Loss: 0.8472 - Val Acc: 69.16%
Epoch [11/25] | Train Loss: 0.6762 - Train Acc: 75.35% | Val Loss: 0.7199 - Val Acc: 71.41%
Epoch [12/25] | Train Loss: 0.6628 - Train Acc: 75.35% | Val Loss: 0.6290 - Val Acc: 76.65%
Epoch [13/25] | Train Loss: 0.6530 - Train Acc: 75.92% | Val Loss: 0.7214 - Val Acc: 72.41%
Epoch [14/25] | Train Loss: 0.6519 - Train Acc: 75.52% | Val Loss: 0.6432 - Val Acc: 76.03%
Epoch [15/25] | Train Loss: 0.6401 - Train Acc: 76.24% | Val Loss: 0.7095 - Val Acc: 72.41%
Epoch [16/25] | Train Loss: 0.6296 - Train Acc: 76.72% | Val Loss: 0.6952 - Val Acc: 74.78%
Epoch [17/25] | Train Loss: 0.6165 - Train Acc: 76.96% | Val Loss: 0.6660 - Val Acc: 77.03%
Epoch [18/25] | Train Loss: 0.6096 - Train Acc: 76.95% | Val Loss: 0.6209 - Val Acc: 77.15%
Epoch [19/25] | Train Loss: 0.5981 - Train Acc: 77.86% | Val Loss: 0.6167 - Val Acc: 77.65%
Epoch [20/25] | Train Loss: 0.5887 - Train Acc: 77.89% | Val Loss: 0.8091 - Val Acc: 70.79%
Epoch [21/25] | Train Loss: 0.5802 - Train Acc: 78.30% | Val Loss: 0.6184 - Val Acc: 78.53%
Epoch [22/25] | Train Loss: 0.5553 - Train Acc: 79.16% | Val Loss: 0.6607 - Val Acc: 74.91%
Epoch [23/25] | Train Loss: 0.5361 - Train Acc: 79.94% | Val Loss: 0.6097 - Val Acc: 77.78%
Epoch [24/25] | Train Loss: 0.5393 - Train Acc: 79.62% | Val Loss: 0.7076 - Val Acc: 75.16%
Epoch [25/25] | Train Loss: 0.5107 - Train Acc: 81.05% | Val Loss: 0.9810 - Val Acc: 63.80%
Best checkpoint saved: best_model/lesions_ResNet18.pth (epoch=21, val_acc=78.53%)
Training Complete!
[sklearn] UndefinedMetricWarning: precision ill-defined (some classes had 0 predicted samples).
```

Final Test Results (ResNet18 on lesions): Accuracy = 76.46% | Precision (macro) = 0.6136 | Recall (macro) = 0.4205 | Macro F1 = 0.4719

Dataset Used – orgs

Organ images (11 classes, grayscale) | Expected minimum test accuracy: 83%

10. orgs – AlexNet

Total epochs trained: 15

Configuration Used:

```
ACTIVATION.................. ReLU
BATCH_SIZE.................. 64
CHANNELS.................... 1
CRITERION................... CrossEntropyLoss
DATA_PATH................... ../data
DEVICE...................... cuda
DROP_RATE................... 0.5
EPOCHS...................... 15
LEARNING_RATE............... 0.001
NUM_CLASSES................. 11
OPTIMIZER................... Adam
RANDOM_SEED................. 42
VAL_SPLIT................... 0.1
WEIGHT_DECAY................ 0
```

Logs:

```
Epoch [01/15] | Train Loss: 0.8939 - Train Acc: 68.87% | Val Loss: 0.4486 - Val Acc: 82.88%
Epoch [02/15] | Train Loss: 0.3669 - Train Acc: 88.23% | Val Loss: 0.4018 - Val Acc: 87.43%
Epoch [03/15] | Train Loss: 0.2964 - Train Acc: 90.30% | Val Loss: 0.4067 - Val Acc: 87.76%
Epoch [04/15] | Train Loss: 0.2334 - Train Acc: 92.26% | Val Loss: 0.1587 - Val Acc: 94.27%
Epoch [05/15] | Train Loss: 0.2069 - Train Acc: 93.15% | Val Loss: 0.2430 - Val Acc: 92.25%
Epoch [06/15] | Train Loss: 0.1724 - Train Acc: 94.26% | Val Loss: 0.2111 - Val Acc: 91.86%
Epoch [07/15] | Train Loss: 0.1439 - Train Acc: 95.13% | Val Loss: 0.1677 - Val Acc: 94.86%
Epoch [08/15] | Train Loss: 0.1488 - Train Acc: 94.99% | Val Loss: 0.1683 - Val Acc: 94.14%
Epoch [09/15] | Train Loss: 0.1360 - Train Acc: 95.57% | Val Loss: 0.3312 - Val Acc: 89.06%
Epoch [10/15] | Train Loss: 0.1337 - Train Acc: 95.65% | Val Loss: 0.0980 - Val Acc: 96.03%
```

```
Epoch [11/15] | Train Loss: 0.1275 - Train Acc: 95.78% | Val Loss: 0.2609 - Val Acc: 94.08%
Epoch [12/15] | Train Loss: 0.1035 - Train Acc: 96.56% | Val Loss: 0.5707 - Val Acc: 87.37%
Epoch [13/15] | Train Loss: 0.1141 - Train Acc: 96.32% | Val Loss: 1.6973 - Val Acc: 77.60%
Epoch [14/15] | Train Loss: 0.1044 - Train Acc: 96.55% | Val Loss: 0.2073 - Val Acc: 92.90%
Epoch [15/15] | Train Loss: 0.1002 - Train Acc: 96.94% | Val Loss: 0.1662 - Val Acc: 94.21%
Best checkpoint saved: best_model/orgs_AlexNet.pth (epoch=10, val_acc=96.03%)
Training Complete!
```

Final Test Results (AlexNet on orgs): Accuracy = 90.07% | Precision (macro) = 0.8894 | Recall (macro) = 0.8856 | Macro F1 = 0.8855

11. orgs – VGG16

Total epochs trained: 15

Configuration Used:

```
ACTIVATION................... ReLU
BATCH_SIZE................... 64
CHANNELS..................... 1
CRITERION.................... CrossEntropyLoss
DATA_PATH.................... ../data
DEVICE....................... cuda
DROP_RATE.................... 0.5
EPOCHS....................... 15
LEARNING_RATE................ 0.001
NUM_CLASSES.................. 11
OPTIMIZER.................... Adam
RANDOM_SEED.................. 42
VAL_SPLIT.................... 0.1
WEIGHT_DECAY................. 0
```

Logs:

```
Epoch [01/15] | Train Loss: 1.3792 - Train Acc: 49.57% | Val Loss: 1.0464 - Val Acc: 61.65%
Epoch [02/15] | Train Loss: 0.6527 - Train Acc: 75.22% | Val Loss: 0.6058 - Val Acc: 78.58%
Epoch [03/15] | Train Loss: 0.5041 - Train Acc: 80.48% | Val Loss: 0.3726 - Val Acc: 84.05%
Epoch [04/15] | Train Loss: 0.4022 - Train Acc: 84.26% | Val Loss: 0.3266 - Val Acc: 86.52%
Epoch [05/15] | Train Loss: 0.3426 - Train Acc: 87.14% | Val Loss: 0.3445 - Val Acc: 86.00%
Epoch [06/15] | Train Loss: 0.3499 - Train Acc: 87.55% | Val Loss: 0.2428 - Val Acc: 90.56%
Epoch [07/15] | Train Loss: 0.2900 - Train Acc: 89.56% | Val Loss: 0.2845 - Val Acc: 89.97%
Epoch [08/15] | Train Loss: 0.2303 - Train Acc: 91.90% | Val Loss: 0.3085 - Val Acc: 89.52%
Epoch [09/15] | Train Loss: 0.2082 - Train Acc: 93.23% | Val Loss: 0.3986 - Val Acc: 88.28%
Epoch [10/15] | Train Loss: 0.3449 - Train Acc: 89.00% | Val Loss: 0.1844 - Val Acc: 93.55%
Epoch [11/15] | Train Loss: 0.1889 - Train Acc: 93.63% | Val Loss: 0.1514 - Val Acc: 94.60%
Epoch [12/15] | Train Loss: 0.1877 - Train Acc: 93.93% | Val Loss: 0.1524 - Val Acc: 94.66%
Epoch [13/15] | Train Loss: 0.1630 - Train Acc: 94.74% | Val Loss: 0.1758 - Val Acc: 94.14%
Epoch [14/15] | Train Loss: 0.1272 - Train Acc: 95.72% | Val Loss: 0.1396 - Val Acc: 95.44%
Epoch [15/15] | Train Loss: 0.1163 - Train Acc: 96.31% | Val Loss: 0.1480 - Val Acc: 94.79%
Best checkpoint saved: best_model/orgs_VGG16.pth (epoch=14, val_acc=95.44%)
Training Complete!
```

Final Test Results (VGG16 on orgs): Accuracy = 88.90% | Precision (macro) = 0.8812 | Recall (macro) = 0.8691 | Macro F1 = 0.8722

12. orgs – ResNet18

Total epochs trained: 15

Configuration Used:

```
ACTIVATION................... ReLU
BATCH_SIZE................... 64
CHANNELS..................... 1
CRITERION.................... CrossEntropyLoss
DATA_PATH.................... ../data
DEVICE....................... cuda
DROP_RATE.................... 0.5
EPOCHS....................... 15
LEARNING_RATE................ 0.001
NUM_CLASSES.................. 11
OPTIMIZER.................... Adam
RANDOM_SEED.................. 42
VAL_SPLIT.................... 0.1
WEIGHT_DECAY................. 0
```

Logs:

```
Epoch [01/15] | Train Loss: 0.6646 - Train Acc: 75.92% | Val Loss: 0.5742 - Val Acc: 81.32%
Epoch [02/15] | Train Loss: 0.3328 - Train Acc: 88.06% | Val Loss: 0.6068 - Val Acc: 80.21%
Epoch [03/15] | Train Loss: 0.2415 - Train Acc: 91.37% | Val Loss: 0.1964 - Val Acc: 92.90%
Epoch [04/15] | Train Loss: 0.1867 - Train Acc: 93.49% | Val Loss: 0.2044 - Val Acc: 92.84%
Epoch [05/15] | Train Loss: 0.1594 - Train Acc: 94.33% | Val Loss: 0.2323 - Val Acc: 91.54%
Epoch [06/15] | Train Loss: 0.1257 - Train Acc: 95.46% | Val Loss: 0.2882 - Val Acc: 91.08%
```

```
Epoch [07/15] | Train Loss: 0.1260 - Train Acc: 95.52% | Val Loss: 0.1832 - Val Acc: 93.95%
Epoch [08/15] | Train Loss: 0.0910 - Train Acc: 96.81% | Val Loss: 0.1936 - Val Acc: 92.90%
Epoch [09/15] | Train Loss: 0.1117 - Train Acc: 96.02% | Val Loss: 0.0998 - Val Acc: 96.16%
Epoch [10/15] | Train Loss: 0.0885 - Train Acc: 97.01% | Val Loss: 0.1554 - Val Acc: 94.99%
Epoch [11/15] | Train Loss: 0.0644 - Train Acc: 97.75% | Val Loss: 0.1134 - Val Acc: 95.96%
Epoch [12/15] | Train Loss: 0.0489 - Train Acc: 98.25% | Val Loss: 0.1877 - Val Acc: 94.08%
Epoch [13/15] | Train Loss: 0.0668 - Train Acc: 97.67% | Val Loss: 0.0999 - Val Acc: 96.09%
Epoch [14/15] | Train Loss: 0.0460 - Train Acc: 98.45% | Val Loss: 0.0729 - Val Acc: 97.14%
Epoch [15/15] | Train Loss: 0.0408 - Train Acc: 98.58% | Val Loss: 0.1739 - Val Acc: 94.79%
Best checkpoint saved: best_model/orgs_ResNet18.pth (epoch=14, val_acc=97.14%)
Training Complete!
```

Final Test Results (ResNet18 on orgs): Accuracy = 92.00% | Precision (macro) = 0.9109 | Recall (macro) = 0.9088 | Macro F1 = 0.9078

Consolidated Summary – All Permutations

Dataset	Model	Accuracy (%)	Precision	Recall	Macro F1
cells	AlexNet	95.15	0.947	0.944	0.945
cells	VGG16	96.78	0.965	0.969	0.967
cells	ResNet18	96.35	0.960	0.964	0.961
chest	AlexNet	86.38	0.905	0.820	0.841
chest	VGG16	82.21	0.889	0.763	0.782
chest	ResNet18	84.46	0.887	0.797	0.817
lesions	AlexNet	76.01	0.565	0.477	0.479
lesions	VGG16	71.22	0.390	0.305	0.324
lesions	ResNet18	76.46	0.614	0.420	0.472
orgs	AlexNet	90.07	0.889	0.886	0.885
orgs	VGG16	88.90	0.881	0.869	0.872
orgs	ResNet18	92.00	0.911	0.909	0.908

Architectural Recommendations

Based on the benchmark results across all twelve dataset model permutations, the recommended pairings are:

Dataset	Recommended model	Test accuracy	Macro F1	Rationale
cells	VGG16	96.78%	0.967	Highest accuracy and F1; AlexNet (95.15%) is a strong lighter-weight alternative when compute is constrained.
chest	AlexNet	86.38%	0.841	Best accuracy and F1 of the three; the shallow architecture generalizes better on this small 2-class set, while the deeper networks overfit.
lesions	ResNet18	76.46%	0.472	Highest accuracy; residual connections help on this hard, imbalanced 7-class task. AlexNet is comparable (76.01%) with a marginally higher F1.
orgs	ResNet18	92.00%	0.908	Best accuracy, precision, recall, and F1 across the board.

Summary: ResNet18 is the strongest general-purpose choice, winning on the two hardest datasets (lesions and orgs) and coming within 0.4% of the best on cells. VGG16 edges out the others only on cells, but at a much higher computational cost. AlexNet is the clear pick for chest and remains a efficient option everywhere, making it the best accuracy per compute trade-off when resources are limited.